



DFDOF

Forensic Baseline Report

Drone Forensic Decision and Orchestration Framework

Case Identifier:	CASE-df048-apr-android-2
Operator:	Floris
Analysis Started:	2026-06-16 11:44:32
Analysis Completed:	2026-06-16 11:46:08
Evidence Directory:	C:\Users\Floris\Documents\dji_drones\train_mavic_air\df048\2018_april\android
Output Directory:	C:\Users\Floris\Documents\dji_drones\train_mavic_air\df048\2018_april\dfdof_output\CASE-df048-apr-android-2
Evidence Sources:	2
Pipeline Version:	DFDOF v1.0 (proof of concept)

This report is an automated forensic baseline produced by DFDOF. It is not a court-ready expert report. All findings are bounded by the data available and must be interpreted by a qualified forensic examiner. The populated case output directory is the authoritative source and provides the full overview of the pipeline's findings.

1. Evidence Intake and Chain of Custody

The following input evidence files were ingested. SHA-256 and SHA-1 hashes were computed at intake to establish integrity baselines. No modification to original evidence files is performed by DFDOF.

File	Classification	Acquisition	Size	Hash Timestamp
android_logical.zip	controller_android	logical	2.3 GB	2026-06-16 11:44:41
df048_internal.E01	drone_sd	physical	7.8 GB	2026-06-16 11:45:07

Evidence Source Path

C:\Users\Floris\Documents\dji_drones\train_mavic_air\df048\2018_april\android

Cryptographic Hashes

android_logical.zip		controller_android	logical
SHA-256:	0c3bd75b0bd8cf8ddd2835dbe303996c3eb2ca6abb0ae8b179791f12e4069d0c		
SHA-1:	a294c910a56a0b92716c510420599252f0cb5d27		

df048_internal.E01		drone_sd	physical
SHA-256:	5664f81579a3558ed7fdf79b06e49cf0849b5f41999ae5a996e43041460ac8f3		
SHA-1:	b9bfc2d90b326950aea2121be9304692b298a288		

2. Pipeline Execution Summary

Completed Phases

1. p1_provenance
2. p2_image_parsing
3. p3_artefact_extraction
4. p4_decision_and_orchestration
5. p5_normalisation_and_anomaly_checking
6. p6_multisource_correlation
7. p7_analysis_and_validation
8. p8_automated_reporting

Anomaly Flags

1. [p1 - drone sd]: mmls: no partition table found in df048_internal.E01; treating as direct partition image
2. [p2 - controller android - device and backup info]: DeviceInfo.xml not found for android_logical.zip
3. [p2 - controller android - device and backup info]: ApplicationInfo.xml not found for android_logical.zip
4. [p2 - controller ios]: source evidence not found
5. [p3 - controller android - account data]: empty file moved to _rejected: dji.go.v4.xml
6. [p3 - controller android - drone logs]: no artefacts found
7. [p3 - controller ios]: source evidence not found
8. [p3 - drone flight storage]: source evidence not found
9. [p4 - controller ios]: source evidence not found
10. [p4 - drone flight storage]: source evidence not found

Tool Execution Status

Tool	Invocations	Successes	Status
exiftool	16	16	[OK]
fls	1	1	[OK]
icat	10	10	[OK]
mmls	1	0	[X] Failed
parser_android	1	1	[OK]
txtlogtocsvtool	1	1	[OK]

Processing Coverage Score

Metric	Result
Evidence Sources Detected	2 / 4
Artefact Categories With Data	5 / 7
Flights With Primary Correlation	0 / 1
Tools Succeeded	5 / 6

3. Device and Account Identification

Field	Value	Confidence	Sources
Installed DJI App(s)	dji.go.v4	[K] Controller-only	1
Drone Serial Number	0K1DF313BD4PR1	[K] Controller-only	1
Drone Model	Mavic Air	[K] Controller-only	1
DJI Application Version	4.2.8	[K] Controller-only	1

4. Flight Analysis - Flight 01

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Flight Record SHA-256: bfd1d4d3bc307f95cb9268ac29e9298f32a39a1f7e540f3438dc5ebc11bc9145
Flight Record Name: DJIFlightRecord_2018-04-19_[11-39-45].csv
Start: 2018-04-19T17:39:46.034000+00:00
End: 2018-04-19T17:41:00.783000+00:00
Duration: 74.7 s

Flight Metrics

Peak Height: 30.2 m
Distance Travelled: 228.99 m
Video Recording: Yes
Photo Capture: No
Photos Taken: 0
Recording Duration: 55.0 s
Record Complete: No - drone had not returned to home point

Fly Modes Observed (Chronological)

1. EngineStart
2. AutoTakeoff
3. GPS_Atti
4. GoHome

Key Events

Timestamp (UTC)	Source	Event	Details
2018-04-19 17:39:46	ctrl_and:fr (v4.2.8)	Log started	SN: 0K1DF313BD4PR1; App: 4.2.8; Mavic Air
2018-04-19 17:39:46	ctrl_and:fr (v4.2.8)	Motor turned on	on
2018-04-19 17:39:46	ctrl_and:fr (v4.2.8)	Record mode changed	No
2018-04-19 17:40:03	ctrl_and:fr (v4.2.8)	Record mode changed	Starting
2018-04-19 17:40:49	ctrl_and:fr (v4.2.8)	Reached peak height	30.2 m
2018-04-19 17:41:00	ctrl_and:fr (v4.2.8)	Log ended	Dist home: 119.7 m; Height: 30.0 m; Rec: 55.0 s

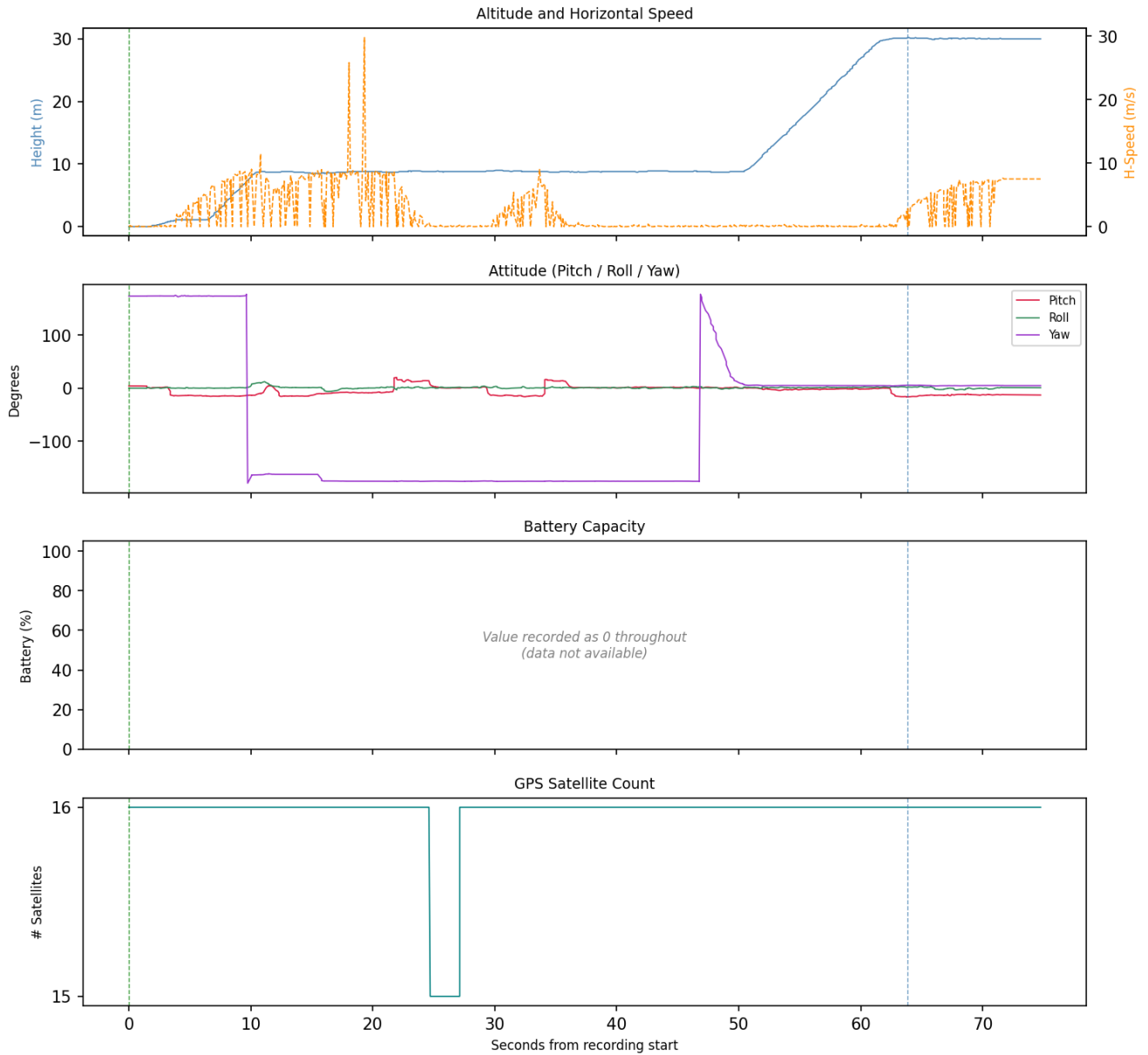
Correlated Artefacts

Possibly correlated (indirect indicators):

controller_android:videos	GPS within flight bounding box (lat 39.9605, lon -106.2166)
p5:d2f1468ffc1d0b2b66b61909c46b9e691b1d78a9416d103cc6e1b5b4907e0cd1	
controller_android: flight_logs	Log messages match (2 entries, format: heading_log)
p5:148cd54f6f474a4378aee89553f1ce783b7811ca8a05863eb4b338491aa40a71	

Flight Record - Sensor Telemetry

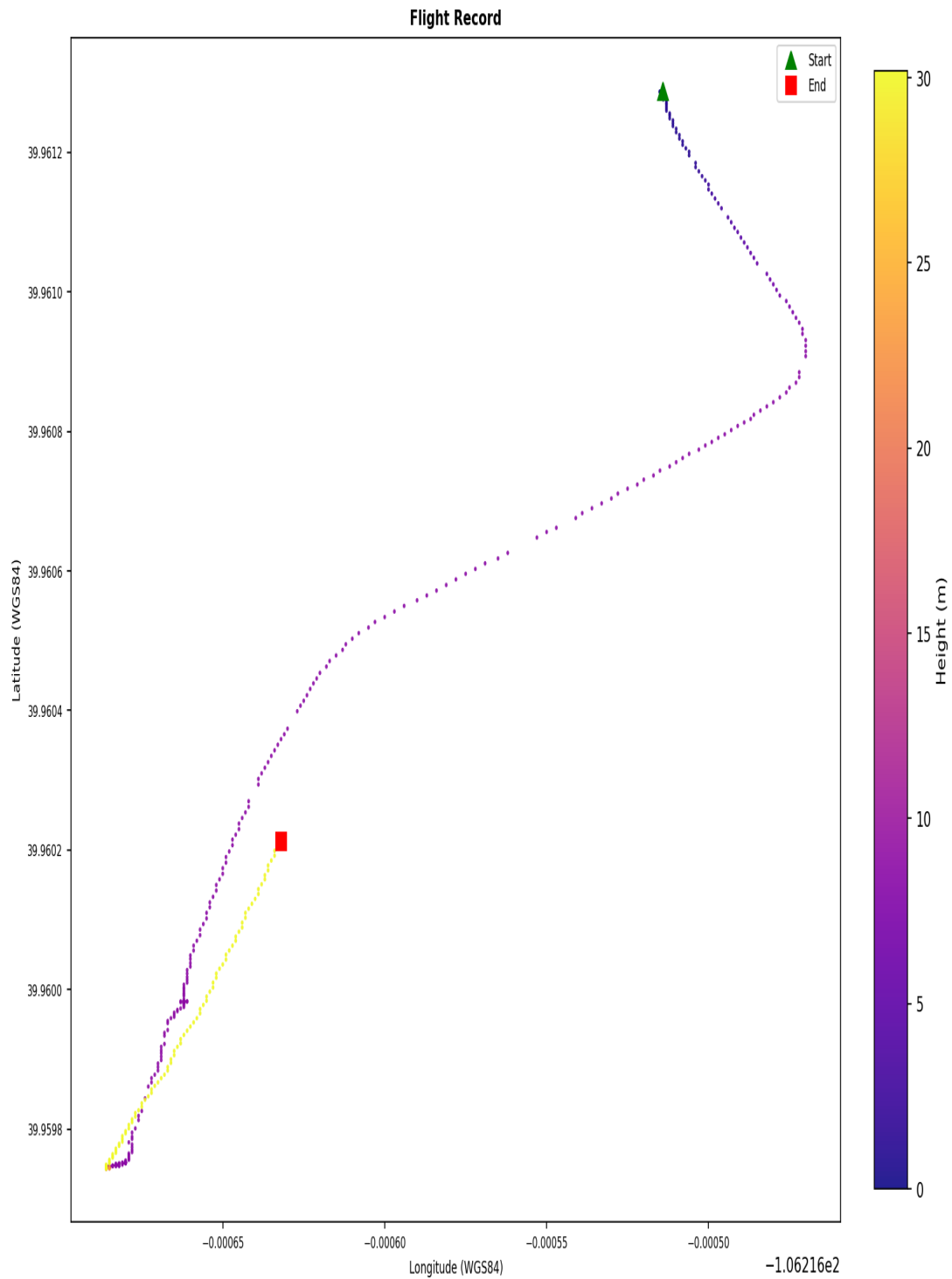
Flight Record Telemetry - flight_01



GPS coordinates are derived from the drone's onboard receiver and represent sensor-recorded positions, not independently verified measurements. A satellite count of 10 or more is associated with good horizontal precision (approximately 1.5-3 m); counts below 4 indicate a poor fix with significantly reduced positional reliability, and those rows are flagged in this report. Low satellite counts may affect the accuracy of the flight path, ground track, and any location-based conclusions drawn from this record.

Flight Ground Track

Flight Ground Track - flight_01



Coordinates are WGS84. No base map. Colour represents relative height (m).

Coordinates are WGS84. No base map. Colour represents relative height (metres).

5. Artefact Coverage

Source Coverage

Source	Detected
controller_android	Yes
controller_ios	No
drone_sd	Yes
drone_flight_storage	No

Artefact Counts per Source and Category

Source	Account Data	Databases	Drone Logs	Flight Logs	Flight Records	Images	Videos
controller_android	0	7	0	5	1	5	2
drone_sd	0	0	0	0	0	5	5

Data Quality Notes per Category

Flight Logs (5 files):

- 3 file(s) unreadable (binary)
- 1 crash dump(s)
- formats: 1 heading_log

Flight Records (1 files):

- 1 file(s) with empty columns
- 1 file(s) with constant-value columns

Images (10 files):

- 10 file(s) missing (normalisable) EXIF timestamp
- 10 file(s) missing EXIF GPS coordinates

Videos (7 files):

- 1 file(s) contain(s) EXIF GPS coordinates
- 6 file(s) missing (normalisable) EXIF timestamp
- 6 file(s) missing EXIF GPS coordinates

6. Data Quality and Anomalies

Data Quality Observations

- Flight_logs: 3 file(s) unreadable (binary) - file content could not be parsed.
- Flight_logs: 1 crash dump(s) - application instability recorded.
- Flight_logs: log files present in the following format(s): 1 heading_log.
- Flight_records: 1 file(s) with empty columns - some telemetry channels contain no data.
- Flight_records: 1 file(s) with constant-value columns - some telemetry channels may be unreliable.

Pipeline Anomaly Flags

1. [p1 - drone sd]: mmls: no partition table found in df048_internal.E01; treating as direct partition image
2. [p2 - controller android - device and backup info]: DeviceInfo.xml not found for android_logical.zip
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9. [p4 - controller ios]: source evidence not found
10. [p4 - drone flight storage]: source evidence not found

Artefact-Level Anomaly Detail

The following summarises which automated checks fired for each artefact processed in Phase 5. Row counts indicate affected data rows; column lists are truncated to five entries.

Flight Logs

19-04-2018-0K1DF313BD4PR1

- format: heading_log | entries: 2

crash-2018-04-22-01-45-10.txt

- format: crash_dump | entries: 1

Flight Records

norm_DJIFlightRecord_2018-04-19_[11-39-45].csv

- contains_no_value: 42 column(s) - APP_SER_WARN.warn, APP_WARN.warn, CENTER_BATTERY.connStatus, CENTER_BATTERY.current [A], CENTER_BATTERY.currentCapacity [mAh] (+37 more)
- contains_constant_value: 175 column(s) - APP_GPS.accuracy, APP_GPS.latitude, APP_GPS.longitude, CAMERA_INFO.cameraType, CAMERA_INFO.connectState (+170 more)

Images

thumb_33177fb665068406_1523618594000.jpg

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

thumb_5be64acbef28816c_1523618190000.jpg

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

thumb_6d9f39f0ca9decc3_1523618188000.jpg

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

thumb_7f92b744bbc53b3a_1523618192000.jpg

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

thumb_dd609e61934788b3_1523618694000.jpg

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0001.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0002.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0003.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0004.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0005.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

Videos**2018_04_19_11_40_02.mp4**

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0001.MP4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0002.MP4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0003.MP4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0004.MP4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0005.MP4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

7. Further Investigation Items

The following items were identified by automated analysis as requiring manual investigator follow-up. They represent unresolved questions or limitations that the pipeline cannot resolve automatically.

1. Note: no controller device name found - Device and account information.
2. Note: no controller device found - Device and account information.
3. Note: no account e-mail found - Device and account information.
4. Note: no product type found - Device and account information.
5. Note: no firmware version found - Device and account information.
6. Note: no last country code found - Device and account information.
7. Note: no aircraft model code found - Device and account information.
8. Note: no controller serial number found - Device and account information.
9. Note: no last known latitude found - Device and account information.
10. Note: no last known longitude found - Device and account information.
11. Note: no account uid found - Device and account information.
12. Note: no account nickname found - Device and account information.
13. Note: no device uuid found - Device and account information.
14. 1 flight record(s) could not be correlated to a drone log (flight_01). Independent corroboration is absent for these flight(s).
15. Flight record(s) show signs of incomplete capture (flight_01). The drone may not have returned to the home point before logging stopped. Manual inspection recommended.
16. 7 database file(s) contain records and should be examined for relevant artefacts.

Appendix A - Extracted Artefact Hash Manifest

Each artefact extracted in Phase 3 is listed below with its full SHA-256 and SHA-1 cryptographic hashes. These values can be used to independently verify the integrity of every extracted file.

dafile.db	Databases	controller_android	extract_logical
SHA-256:	502830ef2494316ccb34c0673b2b9467b6f4d6d6eef931339f79969b5e96042f		
SHA-1:	d60aedd97558fc715ad1386afae948bf4de320f8		

dji.db	Databases	controller_android	extract_logical
SHA-256:	4d4573e8522bd5b40b28e39c8c1836b19d6bf984e6b61ce95afbfaaf6a6ac0b7		
SHA-1:	92c03c60addcb46ced52d314902b2064e841dde		

flysafe_areas_djigo.db	Databases	controller_android	extract_logical
SHA-256:	a21513f2ae3bebe5c80a26c654403e108bef51629a2201a47b8838434abfa70a		
SHA-1:	674c8f1245f5c9c15dc02fffa1ea2690d688f591		

flysafe_areas_djigo_1.db	Databases	controller_android	extract_logical
SHA-256:	d9164be112208833fe06583920459508675742a54e3ca70d358789ab362c7d3c		
SHA-1:	936e7b4a02e38dc7cd2b2d4cb7be8280096b17bd		

flysafe_polygon_1860.db	Databases	controller_android	extract_logical
SHA-256:	bad78ebc14dcc7b63d2b807d0ced2d25d129d8bbc3d93285a6d986f06e0e8cd4		
SHA-1:	a2685bcff9b1ff7e82b4400fbe59b9ca3ff82ebb		

logdb.db	Databases	controller_android	extract_logical
SHA-256:	6b3ec49e43c42ba99961f2f3a29f67f3e3627033f4ae85aef816d4a38f6c879b		
SHA-1:	9c231788699e16000a3a22d34332a287bb85ebd7		

mbgl-offline.db	Databases	controller_android	extract_logical
SHA-256:	fdbdf1daa6d143da25c474eb73402f1f8d429c42362c59a408066ddc5f14ad75		
SHA-1:	4184a13112cae799c637b47d761703130a5f7e58		

1524159485840	Flight Logs	controller_android	extract_logical
SHA-256:	d52123df223056f2e9e081eed457c97522a20845b77a7cb60f0364d7d9e63722		
SHA-1:	39e7073a9e404588475ad25c47883cb4b2f02bed		

1524159534726	Flight Logs	controller_android	extract_logical
SHA-256:	8eb90e0894183be70ba259ca28e46ae76daedb129d87196d7602bb36dc78c32c		
SHA-1:	2e10f8af83ba5fea7bab7bfff2b9d4c710bf2d4f		

1524159661899	Flight Logs	controller_android	extract_logical
SHA-256:	4962a0b5135269a2b06de652d82bc7ed9bc6f93e0718cb340f83d2880c3f17e2		
SHA-1:	8759100f63cc5726698f5cc2864a8b64b321264a		

19-04-2018-0K1DF313BD4PR1	Flight Logs	controller_android	extract_logical
SHA-256:	148cd54f6f474a4378aee89553f1ce783b7811ca8a05863eb4b338491aa40a71		
SHA-1:	1c02a87e58df09a094eb341e2a4884efb62781c9		

crash-2018-04-22-01-45-10.txt	Flight Logs	controller_android	extract_logical
SHA-256:	71380d4bf664e2e5d5a46582502e8c517536f0701f92cac821f8a0ece1b56946		
SHA-1:	5546481f7b623a861a57c6245d88c576515828a1		

DJIFlightRecord_2018-04-19_[11-39-45].txt		Flight Records	controller_android	extract_logical
SHA-256:	cf67262239e3b970b24b8c423b8b2a6890348ff43f7b571de8363a137444cef9			
SHA-1:	7bf87c208a676d3726044b65d118221e8defe4a1			

thumb_33177fb665068406_1523618594000.jpg		Images	controller_android	extract_logical
SHA-256:	e8d6ac3d4cda2bccff4dd3e64950e87fd68700cf80b37f2c45e9e0ac59a4d42a			
SHA-1:	55b6683fe7bf253b609ca32c4fcc3a33795ca21d			

thumb_5be64acbef28816c_1523618190000.jpg		Images	controller_android	extract_logical
SHA-256:	e6a060c7756c7e2fd99c850434b02c708a329ebf16a814e0fe0a77b15caaab24			
SHA-1:	ed1bfdceab3e02d87575cadd6c79fdd983021271			

thumb_6d9f39f0ca9decc3_1523618188000.jpg		Images	controller_android	extract_logical
SHA-256:	cbe6209aa64bbb44340e4aed5d2e276fd1df75805f956307b24a83e79b1d6f90			
SHA-1:	5b90ca57716f15045a6a65c7215ca77036ff6af2			

thumb_7f92b744bbc53b3a_1523618192000.jpg		Images	controller_android	extract_logical
SHA-256:	32ff09733d14f4ecdd59171811fd50111cc00752ebdb528f04b75b08cf220874			
SHA-1:	874dff84aaa18baf06b4070276135346c328e178			

thumb_dd609e61934788b3_1523618694000.jpg		Images	controller_android	extract_logical
SHA-256:	8986d1b4a162ca4bbe0adb12f01141b9d7673ba196436ee31e16fb6b540b2e2			
SHA-1:	3802402288db8466063ecaec81012031de544392			

DJI_0001.THM		Images	drone_sd	extract_physical
SHA-256:	3a3918e0c00dddb9ad706491020da1b7db90479f49e29814274087259da2bcf0			
SHA-1:	b6b5af5e4d4b0d5ecae0eef6e7901ec873ce8048			

DJI_0002.THM		Images	drone_sd	extract_physical
SHA-256:	c452f70bc9e9bad9f05047d07cf685498ab5117ec940a5bc6622bd77baa79b1f			
SHA-1:	9f15bf98c1c26619f2499ece191f0dc426aae6e3			

DJI_0003.THM		Images	drone_sd	extract_physical
SHA-256:	9727e9310aa901a379b5fdfaf29a50e22fb879ff86d7ba6d75e82c6a39497d79			
SHA-1:	c958de01cff11f28d30a028a327b0d9867c968fc			

DJI_0004.THM		Images	drone_sd	extract_physical
SHA-256:	f99c4b6f1f614bd2ea9829aa50e08d966ed188dde73ce9dfcf388ee70adb7dc4			
SHA-1:	434ae634442c15bfbe0b51dbc2b31193f14968d5			

DJI_0005.THM		Images	drone_sd	extract_physical
SHA-256:	b3f920d7ad5412a19b8f122aaa0ff4cedfcf9b25c2baa89bbf4d26e0d796fe5f			
SHA-1:	6f4b46f652887ad89b44020dc176a4ac9f0bb324			

2018_04_19_11_40_02.info		Videos	controller_android	extract_logical
SHA-256:	d2f1468ffc1d0b2b66b61909c46b9e691b1d78a9416d103cc6e1b5b4907e0cd1			
SHA-1:	8d45b0a1e17a07551a4315a249425f66813aba90			

2018_04_19_11_40_02.mp4		Videos	controller_android	extract_logical
SHA-256:	478d1d59aed00f642ceabca80e3c63d994a258e450b2007ce79faac4714aa165			
SHA-1:	1fc46d9b4740f6b7190cc1315cbcdacd01e3a779			

DJI_0001.MP4		Videos	drone_sd	extract_physical
SHA-256:	61a290fa102ce3e2913242b83aae8073af3751cd6b8f1ed0e14eab7625f77815			
SHA-1:	9873e6568f8631e23ae819daa214f6e88312efea			

DJI_0002.MP4		Videos	drone_sd	extract_physical
SHA-256:	9db5019a6c22e8ce2c2d726df4f8c1c0b52fc3d5806b38058bf1896f80e6ba7b			
SHA-1:	5131359773d614150ff1d62d8ee4c4d984eb640f			

DJI_0003.MP4		Videos	drone_sd	extract_physical
SHA-256:	aad026be3d81c7dbf251c37d7be4776d239dd44cd7a9988190e6a0f04b87796a			
SHA-1:	e74bf86c588742fb6cf85fda3b641c41e7834b65			

DJI_0004.MP4		Videos	drone_sd	extract_physical
SHA-256:	6aa7dc8ae4f82d9238d99673a3b4c30ab87f3dfe46627de30b2d6f200beacb3			
SHA-1:	8a6bfb4eca3908e47a7bab171aa907e333ed0ab7			

DJI_0005.MP4		Videos	drone_sd	extract_physical
SHA-256:	890e7c07f2d747cdbdffe8939636fc1c4f1230ce592db8d91fff8742551f8192			
SHA-1:	a262dd9af22576b92269280cc9e1fe567e375821			

Appendix B - Evidence Derivation Summary

This table summarises how many artefacts were derived at each pipeline phase, per evidence source and category. It demonstrates the processing chain from extracted artefacts (P3) through decoded outputs (P4) to normalised analysis-ready files (P5).

Controller Android

Category	P3 Extracted	P4 Decoded	P5 Normalised
Databases	7	0	0
Flight Logs	5	0	0
Flight Records	1	1	1
Images	5	5	0
Videos	2	2	0

Drone Sd

Category	P3 Extracted	P4 Decoded	P5 Normalised
Images	5	5	5
Videos	5	5	0

Appendix C - Tool Invocation Log

All external tool invocations executed during this analysis are recorded below, in chronological order. Each entry includes the timestamp, tool name, version, return code, full command, and output files produced.

2026-06-16 11:44:33 mmls vThe Sleuth Kit ver 4.15.0 rc=1

cmd: mmls.exe df048_internal.E01

2026-06-16 11:44:33 fls vThe Sleuth Kit ver 4.15.0 rc=0

cmd: fls.exe -r -p df048_internal.E01

2026-06-16 11:45:07 parser_android v- rc=n/a

cmd: android_logical.zip C:\Users\Floris\Documents\dfdof_output\CASE-df048-apr-android-2\p2_image_parsing\controller_android_parsed
out: controller_android_parsed

2026-06-16 11:45:12 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_internal.E01 1357829
out: DJI_0001.MP4

2026-06-16 11:45:17 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_internal.E01 1357830
out: DJI_0002.MP4

2026-06-16 11:45:31 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_internal.E01 1357831
out: DJI_0003.MP4

2026-06-16 11:45:48 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_internal.E01 1357832
out: DJI_0004.MP4

2026-06-16 11:45:58 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_internal.E01 1357833
out: DJI_0005.MP4

2026-06-16 11:46:02 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_internal.E01 1358853
out: DJI_0001.THM

2026-06-16 11:46:02 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_internal.E01 1358854
out: DJI_0002.THM

2026-06-16 11:46:02 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_internal.E01 1358855
out: DJI_0003.THM

2026-06-16 11:46:03 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_internal.E01 1358856
out: DJI_0004.THM

2026-06-16 11:46:03 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_internal.E01 1358857
out: DJI_0005.THM

2026-06-16 11:46:03 txtlogtocsvtool v2018-06-11 rc=0

cmd: TXTlogToCSVtool.exe DJIFlightRecord_2018-04-19_[11-39-45].txt DJIFlightRecord_2018-04-19_[11-39-45].csv
out: DJIFlightRecord_2018-04-19_[11-39-45].csv

2026-06-16 11:46:03 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n thumb_33177fb665068406_1523618594000.jpg
out: thumb_33177fb665068406_1523618594000_exif.json

2026-06-16 11:46:04 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n thumb_5be64acbef28816c_1523618190000.jpg
out: thumb_5be64acbef28816c_1523618190000_exif.json

2026-06-16 11:46:04 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n thumb_6d9f39f0ca9decc3_1523618188000.jpg
out: thumb_6d9f39f0ca9decc3_1523618188000_exif.json

2026-06-16 11:46:04 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n thumb_7f92b744bbc53b3a_1523618192000.jpg
out: thumb_7f92b744bbc53b3a_1523618192000_exif.json

2026-06-16 11:46:04 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n thumb_dd609e61934788b3_1523618694000.jpg
out: thumb_dd609e61934788b3_1523618694000_exif.json

2026-06-16 11:46:05 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2018_04_19_11_40_02.mp4
out: 2018_04_19_11_40_02_exif.json

2026-06-16 11:46:05 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0001.THM
out: DJI_0001_exif.json

2026-06-16 11:46:05 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0002.THM
out: DJI_0002_exif.json

2026-06-16 11:46:05 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0003.THM
out: DJI_0003_exif.json

2026-06-16 11:46:06 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0004.THM
out: DJI_0004_exif.json

2026-06-16 11:46:06 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0005.THM
out: DJI_0005_exif.json

2026-06-16 11:46:06 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0001.MP4
out: DJI_0001_exif.json

2026-06-16 11:46:07 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0002.MP4
out: DJI_0002_exif.json

2026-06-16 11:46:07 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0003.MP4
out: DJI_0003_exif.json

2026-06-16 11:46:07 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0004.MP4
out: DJI_0004_exif.json

2026-06-16 11:46:08 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0005.MP4
out: DJI_0005_exif.json

Appendix D - DFDOF Proof of Concept Overview

The Drone Forensic Decision and Orchestration Framework (DFDOF) is an automated 8-phase forensic pipeline for DJI drone evidence. It ingests images of the controller, drone flight storage and SD cards, then processes them into a structured analysis and report.

P1 Provenance and Integrity

Classifies input files (ZIP/E01/.001) by content signatures. Computes SHA-256 and SHA-1 for all input evidence. Presents a source classification summary for operator confirmation before processing begins.

Input: evidence directory. Output: classified evidence objects with hashes.

P2 Image Parsing

Makes all sources search-ready. Decrypts iOS backups into a domain directory tree. Extracts Android device metadata (DeviceInfo.xml, ApplicationInfo.xml, packages.list). Produces backup_info.json with device identifiers and installed DJI applications.

Input: raw evidence files. Output: parsed directory structures, backup_info.json.

P3 Artefact Extraction

Root-driven category extraction. Discovers DJI app roots by bundle-ID segment matching. Recurses into configured per-platform category paths and filters by extension. Rejects empty files. Each artefact is wrapped in an Evidence object with a parent hash linking it to its source.

Input: parsed directories. Output: categorised artefacts (flight records, drone logs, flight logs, databases, images, videos, account data).

P4 Decision and Orchestration

Dispatches each artefact category to the appropriate specialist tool: DatCon for .DAT drone logs (produces CSV + KML), TXTlogToCSV for flight record .txt files, ExifTool for images and videos, Python parsers for account data (.plist/.xml). Computes hashes for all tool outputs.

Input: extracted artefacts. Output: decoded CSVs, KML files, EXIF JSON, account data observations.

P5 Normalisation and Anomaly Checking

Normalises all CSVs to UTC timestamps and a unified schema. Applies 9 shared telemetry anomaly checks (timestamp regression/gaps, zero coordinates, GPS quality, altitude spikes, motor-airborne-off) plus format-specific checks for flight records (battery voltage, temperature) and drone logs (quaternion, hDOP). Detects empty databases, unreadable flight logs, and missing (normalisable) EXIF timestamps.

Input: decoded artefacts. Output: normalised CSVs, anomaly observations.

P6 Multi-Source Correlation

Correlates flight records with drone logs using two criteria: (1) temporal overlap ≥ 60 s AND $\geq 90\%$ of the shorter flight's duration; (2) median haversine distance of nearest-neighbour GPS pairs ≤ 25 m with ≥ 5 matched pairs. Deduplicates FR candidates when multiple apps recorded the same flight. Builds a unified per-flight event timeline from all matched sources. Also correlates media files by EXIF timestamp, GPS bounding box, and video duration. In the correlation summary: Overlap is the temporal overlap as a percentage of the shorter flight's duration; GPS Match Rate is the fraction of flight-record GPS rows that found a spatially matched drone-log row within a ± 2 s window.

Input: normalised CSVs. Output: timeline_flightXX.json per flight.

P7 Analysis and Validation

Aggregates results from all prior phases into a structured analysis. Resolves device identity (drone serial, model, app version, account email) across sources with confidence levels (controller-only / corroborated / inconsistent). Computes per-flight metrics (peak height, media capture, record completeness). Produces coverage scores and a grouped conclusions structure with a dedicated further_investigation section.

Input: all phase outputs. Output: analysis dict with conclusions in state.json.

P8 Automated Reporting

Generates this PDF forensic baseline report from all prior phase outputs. Produces sensor telemetry plots (altitude, speed, attitude, battery, GPS) from normalised CSVs, a dual-source flight track plot, and the structured appendices. All figures are also saved as standalone PNG files.

Input: state.json (all phase outputs). Output: PDF report + PNG plot files.

Disclaimer

DFDOF was developed by Floris Krijger, MSc Security and Network Engineering, University of Amsterdam, as a Master's thesis project (1 April 2026 - 30 June 2026). This proof of concept has been developed and tested on a targeted subset of the NIST VTO forensic drone dataset and is not a finished, production-ready system. Before use in any operational or legal context, further testing, validation across additional drone models and evidence types, and robustness improvements are required. The authors accept no liability for conclusions drawn from automated pipeline output without independent expert verification.